

Skills Summary

Counting a Set and Telling How Many

Use this reference while completing your worksheet or when studying.

The counting rule (one-to-one). Touch *one* object, say *one* number. Touch every object, and touch each one only once.

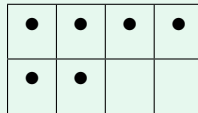
The how-many rule (cardinality). The **last number you say** tells how many are in the whole set. You do not have to count again to answer “how many?”

Helpers: a full row of a ten-frame is 5; a full ten-frame is 10.

1. Count a set and tell how many

Give the set an order before you start — left to right, top row then bottom row — and cross off or slide each object as you count it. Order is what keeps you from skipping one or counting one twice.

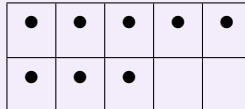
Example: How many dots?



Step 1. The dots sit in two rows, so count the full row first and then count on into the second row instead of starting over.

Step 2. Top row: one, two, three, four — that is 4. Count on: “five, six”, which adds 2. The last number said is the answer: **6** dots.

Try it: How many dots?



check: 8

2. Count two groups as one set

When a set is shown in two places — two baskets, two colours, two rows — count the first group, then **keep going** into the second group. Do not start again at one. The last number you say is how many there are *in all*.

Example: 3 red cubes and 4 blue cubes. How many cubes in all?

Step 1. Both colours are cubes in the same pile, so one count has to cover both groups.

Step 2. Count the red ones: one, two, three. Now keep counting the blue ones: “four, five, six, seven”. That is $3 + 4$, so there are **7** cubes in all.

Try it: 6 pencils in the cup and 5 pencils on the desk. How many pencils in all?

check: 11

3. Tell which set has more

Finish **both** counts first, then compare the two numbers.

$>$ means *is greater than* $<$ means *is less than*

The open end of the sign always faces the bigger number.

Example: Ana has 4 stickers and then 2 more. Ben has 3 stickers and then 2 more. Who has more?

Step 1. Nothing can be compared until each child's set has one number, so count each set all the way first.

Step 2. Ana: $4 + 2 = 6$. Ben: $3 + 2 = 5$. Six is more than five. Ana's set is greater, so Ana $>$ Ben.

Try it: Kim has 5 marbles and then 2 more. Lee has 5 marbles and then 4 more. Write $>$ or $<$: Kim ____ Lee.

$>$:check

(!) Watch out: a bigger-looking set is not always more — spread-out objects can be fewer than crowded ones, and only the count decides. And when someone asks “how many?” after you have counted, say the last number again. Recounting the set is the sign that the how-many rule has not clicked yet.